

Profile Likelihood-Ratio Intervals, the Bartlett Correction, and the Eigenvalue-Spectrum Reference Law

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Purpose

Records the theory behind the experiment-43 maximal-reliability interval: the profile likelihood-ratio (LR) statistic and its inversion into a confidence set, the Bartlett correction that rescues its small- N calibration under normality, and what changes under non-normality, where the reference law is a weighted sum of χ_1^2 variables (the full eigenvalue spectrum, of which Satorra-Bentler scaling is only the first-moment summary). Three claims, stated here and argued below.

1. Test inversion buys transformation-invariance, range-respecting limits, and asymmetry for free, by anchoring on the likelihood rather than on $\hat{\psi}$ (Section 1).
2. The Bartlett correction is a finite-sample ($O(N^{-1})$) fix that is valid *only when the limit is genuinely χ^2* . It removes the small- N inflation of the LR and, uniquely among the trinity, does so to one extra order (Section 2).
3. Under non-normality the leading-order ($O(1)$) error must be fixed first by a spectral correction (Section 3). A single scalar Bartlett factor on the mixture is then only a first-moment patch, not a Bartlett correction in the Barndorff-Nielsen sense; the principled finite-sample-robust route is to correct (or resample) the eigenvalues, not to rescale a mean (Section 4). For a *one*-parameter interval the spectrum collapses to a single eigenvalue, so there the robust correction is just a scalar and the distinction is moot; it bites only for $\text{df} > 1$ (overall fit, multi-parameter regions, nested tests).

1 The profile LR and test-inversion intervals

Partition $\theta = (\psi, \lambda)$ with $\psi \in \mathbb{R}$ the interest parameter (here the maximal reliability ρ^*) and λ the nuisance block. In covariance-structure ML the discrepancy F satisfies $-2\ell(\theta)/N = F(\theta) + \text{const}$, and we write $\hat{\theta} = \arg \min_{\theta} F$ for the unconstrained fit and $\tilde{\theta}(\psi_0) = \arg \min_{\theta: \psi=\psi_0} F$ for the fit constrained to $\psi = \psi_0$. The profile discrepancy is $F_p(\psi) = \min_{\lambda} F(\psi, \lambda)$, and the profile LR statistic is

$$W(\psi_0) = N[F_p(\psi_0) - F_p(\hat{\psi})] = N[F(\tilde{\theta}(\psi_0)) - F(\hat{\theta})], \quad (1)$$

which is exactly the experiment-43 statistic $\text{LR} = N(F_{\text{con}} - F_{\text{unc}})$.

Theorem 1 (Wilks). *Under the usual regularity conditions, correct specification, normal-theory ML, and θ interior with true interest value ψ_0 , $W(\psi_0) \xrightarrow{d} \chi_q^2$ with $q = \dim \psi$ ($= 1$ here) [1].*

The $(1 - \alpha)$ confidence set is the acceptance region of the level- α LR test, swept over the hypothesized value:

$$C_{1-\alpha} = \{\psi_0 : W(\psi_0) \leq \chi_{q, 1-\alpha}^2\}. \quad (2)$$

Coverage of the true value needs only W at that value, which is why experiment-43 measures coverage with one constrained fit per replicate and defers interval width (the latter needs the root-finding $W(\psi_0) = \chi_{q, 1-\alpha}^2$).

Four properties make (2) succeed where intervals built around $\hat{\psi}$ fail.

- (i) *Transformation-invariance.* W is invariant under any monotone reparameterization of ψ ; a reparameterization maps C to the image set. No logit-versus-Fisher choice arises.
- (ii) *Range-respecting.* The constrained fit stays in the parameter space, so $C \subseteq [0, 1]$ for a reliability by construction; the interval cannot leave the natural range.
- (iii) *Asymmetry.* C follows the curvature of F_p and is asymmetric whenever the profile likelihood is.
- (iv) *No standard error.* Only fits enter, never an estimate of $\text{Var}(\hat{\psi})$, so C is immune to the bias, skew, and boundary pileup of $\hat{\psi}$ that defeat Wald, delta, and Cornish–Fisher intervals.

Remark. W , the Wald statistic $W_W = (\hat{\psi} - \psi_0)^2 / \widehat{\text{Var}}(\hat{\psi})$, and the score statistic are asymptotically equivalent to $o_p(1)$ and share the limit of Theorem 1. W_W is the quadratic Taylor approximation of W at $\hat{\psi}$; it is *not* transformation-invariant and inherits every finite-sample defect of $\hat{\psi}$. The signed root $R(\psi_0) = \text{sign}(\hat{\psi} - \psi_0) \sqrt{W(\psi_0)} \rightarrow N(0, 1)$ is the one-sided handle and the entry point for higher-order (r^*) refinements. In SEM, profile-LR intervals are due to [2] (genetics, the OpenMx `mxCI` machinery) and [3] (general SEM, with confidence regions); the R package `sem1bci` implements them.

2 The Bartlett correction

Theorem 1 is leading-order only. The next term is a pure scale effect:

$$\mathbb{E}[W(\psi_0)] = q \left(1 + \frac{b}{N} + O(N^{-2}) \right), \quad (3)$$

with b a model-dependent constant. At small N the mean exceeds q , W is stochastically too large, the test over-rejects, and C under-covers. In experiment-43 this is stark: $\mathbb{E}[W] \approx 3.8$ against $q = 1$ at $N = 50$ for the general-factor coefficient, decaying to ≈ 1 by $N = 200$.

Theorem 2 (Bartlett-correctability). *Let $c = 1 + b/N = \mathbb{E}[W]/q$ and $W' = W/c$. Then $W' \xrightarrow{d} \chi_q^2$ with relative error $O(N^{-2})$, one order better than for W [4, 5]. The single mean-matching rescaling that sets $\mathbb{E}[W'] = q$ simultaneously corrects the higher cumulants to the order needed; the level error after adjustment is $O(N^{-2})$ [6, 7]. This is special to the LR: the Wald and score statistics are not Bartlett-correctable by a scalar.*

The correction is the rescaling $W \mapsto W/c$ in (2), equivalently a rescaled threshold $c \chi_{q,1-\alpha}^2$. The factor $c = \mathbb{E}[W]/q$ can be obtained three ways:

- analytically, from Lawley's general expansion of (3) (model-specific and algebraically heavy);
- by parametric bootstrap, simulating W under the fitted null and taking the sample mean;
- empirically/oracle, the Monte-Carlo mean of W at the true value, as in experiment-43 (an oracle, not an operational estimator).

Remark (SEM lineage). The covariance-structure analogue is old. [8] gave the factor-analysis χ^2 multiplier; [9] the Swain correction (a Bartlett-type

scalar for the ML χ^2 , evaluated by [10, 11]); [12] an empirical correction tuned for many variables. All target the same $O(N^{-1})$ over-rejection of the normal-theory χ^2 . The standing assumption everywhere is that the limit *is* χ^2 : a Bartlett correction fixes the finite-sample approach to that limit and presupposes it exists.

3 The reference law under non-normality: the full spectrum

Under non-normality (or misspecification) the normal-theory statistic no longer converges to χ^2 . Let $\Gamma = \text{Cov}(\sqrt{N} \text{vech}(S - \Sigma))$ be the asymptotic covariance of the sample moments, $\Delta = \partial \text{vech} \Sigma(\theta) / \partial \theta'$ the model Jacobian, V the fit weight (normal-theory $V = \Gamma_{\text{NT}}^{-1}$ for ML), and

$$U = V - V\Delta(\Delta'V\Delta)^{-1}\Delta'V \quad (4)$$

the residual-weight (projection) matrix.

Theorem 3 (Spectral limit). *Under correct structural specification and finite fourth moments,*

$$T = N\hat{F} \xrightarrow{d} \sum_{j=1}^r \lambda_j z_j^2, \quad z_j \stackrel{iid}{\sim} N(0, 1), \quad (5)$$

where $\lambda_1, \dots, \lambda_r$ are the $r = \text{df}$ nonzero eigenvalues of $U\Gamma$ [13, 14, 15]. Under multivariate normality $\Gamma = \Gamma_{\text{NT}}$ and $U\Gamma_{\text{NT}}$ is idempotent of rank r , so every $\lambda_j = 1$ and $T \rightarrow \chi_r^2$. Non-normality moves the eigenvalues off 1.

So the statistic is a weighted sum of independent χ_1^2 , not a χ^2 . The established corrections form a hierarchy by how much of the spectrum $\{\lambda_j\}$ they use:

- *Mean* (Satorra–Bentler T_M): scale by $c_1 = \text{tr}(U\Gamma)/r = \bar{\lambda}$ and refer to χ_r^2 . First moment only; the variance is wrong unless the λ_j are equal.
- *Mean-and-variance* (Satorra–Bentler adjusted / Satterthwaite T_{MV}): match two moments by scaling and setting fractional $d = (\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$; refer to χ_d^2 .
- *Scaled-shifted* (T_{MV2}): match two moments by an affine $aT + b$ at fixed df [16].

- *Higher moments*: third-moment-adjusted variants extend the same idea.
- *Full spectrum*: use all λ_j and the exact distribution of $\sum_j \lambda_j z_j^2$ by numerical inversion of its characteristic function [17, 18], with no moment truncation.

Remark (Why the full spectrum is hard: the FMG program). The exact tail of (5) needs the population λ_j , but the sample eigenvalues of $\hat{U}\hat{U}^T$ are badly biased and unstable, worst for the small eigenvalues and exactly where N/df is small; plugging them into Imhof is then worse than moment-matching. Eigenvalue block averaging (EBA) bins the ordered sample eigenvalues and replaces each by its block mean, trading variance for bias [19]; penalized EBA (pEBA / pOLS) shrinks the spectrum toward a model-based trend and holds the nominal rate where T_M and T_{MV} drift under non-normality [20], with the nested-model extension in [21]. magmaan implements this family (experiments 17–18) to $\sim 3 \times 10^{-6}$ parity with `semTests`. Satorra–Bentler is, in this language, the rank-collapsed first-moment summary of the same spectrum the full method resolves.

4 A Bartlett correction for the spectral statistic?

The two corrections attack different orders and are orthogonal. The spectral correction fixes the leading-order ($O(1)$) reference law: it makes the *asymptotic* distribution right under non-normality. The Bartlett correction fixes the next-order ($O(N^{-1})$) approach to a χ^2 limit. Under non-normal, small- N data *both* are wrong, and one needs both. Three things “Bartlett the spectrum” can mean.

(a) A first-moment finite-sample patch (exists; recommended).

Apply a Bartlett-type scalar to an *already-robust* base statistic: T_{MB} = Bartlett-corrected Satorra–Bentler. [22] compare the family and recommend exactly this ordering, mean-correct (SB) first, then a Bartlett scalar on the scaled statistic, reporting that T_{MB} is satisfactory across sources of non-normality where bare SB and the moment-matched variants drift at small N . This is the operational answer and the direct sibling of experiment-43, with the SB mean replacing the normal-theory mean in the factor.

(b) A true Bartlett correction of the mixture (does not exist as a scalar).

The Barndorff-Nielsen property of Theorem 2, that one scalar fixes

the whole distribution to $O(N^{-2})$, is a property of the LR with a χ^2 limit. For a general weighted- χ^2 limit there is no single scalar whose mean-match also corrects the higher cumulants: the finite-sample error lives in the whole eigenvalue vector (each λ_j carries its own $O(N^{-1})$ bias), not in one mean. Properly “Bartlett-correcting the spectrum” therefore means finite-sample-correcting the eigenvalues (or Γ and U) before Imhof, which is precisely the bias problem EBA/pEBA already attack by shrinkage; pEBA is, in effect, the spectral analogue of a finite-sample correction, done by penalization rather than by an $O(N^{-1})$ expansion.

(c) Resample the reference law (the clean robust-plus-finite-sample route). A model-based or nonparametric bootstrap [23] estimates the finite-sample null distribution of T directly, absorbing non-normality (leading order) and the $O(N^{-1})$ term at once; the bootstrap mean of T is a Bartlett factor that is robust by construction. This is why the empirical c of experiment-43 is more than a normal-theory Bartlett factor: re-estimated on non-normal data it would absorb the leading-order mis-scaling too. A double bootstrap refines the calibration further.

Remark (The $\text{df}=1$ collapse). For a one-parameter interval ($q = 1$), the constrained-LR limit under non-normality is a *single* scaled variable λz^2 : the spectrum of the relevant $q \times q$ matrix has one element. There the robust correction is just the scalar scaling λ (a Satorra-2000 scaled difference test), the “full spectrum” has nothing extra to offer, and T_{MB} of path (a) is the complete recipe. The spectral richness of Section 3 matters only for $\text{df} > 1$: overall goodness-of-fit, multi-parameter confidence regions, and nested tests with more than one restriction.

Recommendation for a built surface (experiment 43 / `measures::frontier`).

- Normal data: profile-LR interval (2) with a Bartlett factor (analytic or parametric-bootstrap c). A genuine Bartlett correction, $O(N^{-2})$.
- Non-normal data: do *not* Bartlett the normal-theory LR. Either (i) scale by the robust λ (Satorra-2000) then Bartlett (T_{MB} , path (a)), or (ii) bootstrap the profile-LR null (path (c)). Both fix leading order first.
- The eigenvalue machinery already in magmaan (Imhof tail, EBA/pEBA) is the spectral reference; the open piece is wiring a spectral (not χ_1^2) threshold into the profile-LR inversion, the non-normal sibling that the experiment-43 caveat defers.

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